Jorge Carlos Valverde-Rebaza

Professor of Computer Science

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Summary

- Professor and researcher in computer science, data analytics, and artificial intelligence, with experience in academia and industry across Mexico, Brazil, Peru, and Europe.

Education

Qualification — **Consejo Nacional de Ciencia y Tecnología e Innovación Tecnológica (CONCYTEC)**, Investigador RENACYT (Grupo Carlos Monge Medrano - Nivel III)
- CÓDIGO RENACYT: P0005495

Lima, Peru
Oct 2019 – Oct 2021

PhD — **Universidade de São Paulo**, Doctor in Computer Science and Computational Mathematics
- Instituto de Ciências Matemáticas e de Computação

Sao Paulo, SP, Brazil
Jan 2013 – Jan 2017

Research Visit — **Université de Montpellier**, Scientific Visitant
- Montpellier Laboratory of Informatics, Robotics and Microelectronics (LIRMM)

Montpellier, France
Jan 2015 – Jan 2015

Research Visit — **Universidade do Porto**, Scientific Visitant
- Laboratory of Artificial Intelligence and Decision Support (LIAAD)

Porto, Portugal
Jan 2014 – Jan 2014

MSc — **Universidade de São Paulo**, Master in Computer Science and Computational Mathematics
- Instituto de Ciências Matemáticas e de Computação

Sao Paulo, SP, Brazil
Jan 2011 – Jan 2013

Bachelor — **Universidad Nacional de Trujillo**, Bachelor in Computer Science
- Facultad de Ciencias Físicas y Matemáticas

Trujillo, La Libertad, Peru
Jan 2004 – Jan 2008

Experience

Tecnológico de Monterrey, Professor, School of Engineering and Science
Professor at the School of Engineering and Science, involved in teaching and research in data analytics, artificial intelligence, and related areas.

Mexico City, DF, Mexico
July 2020 – present
6 years 5 months

Visibilia, Chief Research Scientist
Chief Research Scientist at the Department of Scientific Research, leading projects in data science, AI, and business intelligence.

São Paulo, SP, Brazil
Sept 2017 – July 2024
6 years 11 months

Faculdade Impacta, Professor
Professor in computer science–related courses.

São Paulo, SP, Brazil
Feb 2020 – June 2020
6 months

Universidad Peruana de Ciencias Aplicadas, Associate Professor
Associate Professor in computer science and data analytics.

Monterrico, Lima, Peru
Mar 2018 – July 2019
1 year 6 months

Esan University, Professor
Professor in computer science and related disciplines.

Monterrico, Lima, Peru
Feb 2018 – Dec 2018
11 months

Projects

Generative AI as a tool for building advanced data analytic solutions for students without programming skills
Grant, Institute for the Future of Education (Monterrey, MX), 2024-01 to 2025-07. GRANT_NUMBER: 419283.

Digital Media Observatory
Grant, Vice-Rectorate for Research at Tecnologico de Monterrey (Monterrey, MX), 2023-01 to 2025-12. URL: https://omd.tec.mx/

BEENET: The solution to the problem of offer and seek employment
Grant, Fundacao de Amparo a Pesquisa do Estado de Sao Paulo (Sao Paulo, BR), 2019-11 to 2021-10. GRANT_NUMBER: 2019/02455-1. URL: https://bv.fapesp.br/pt/auxilios/105710/beenet-a-solucao-para-o-problema-da-oferta-e-procura-de-emprego/

An Intelligent System Proposal for Identifying Business Opportunities
Grant, Fundacao de Amparo a Pesquisa do Estado de Sao Paulo (Sao Paulo, BR), 2018-11 to 2019-07. GRANT_NUMBER: 2017/22472-2. URL: https://bv.fapesp.br/pt/auxilios/101839/desenvolvimento-de-um-sistema-inteligente-para-identificacao-de-oportunidades-de-negocios/

Mining User Behavior in Location-Based Social Networks
Grant, Fundacao de Amparo a Pesquisa do Estado de Sao Paulo (Sao Paulo, BR), 2014-03 to 2017-09. GRANT_NUMBER: 2013/12191-5. URL: https://bv.fapesp.br/pt/bolsas/150470/mineracao-do-comportamento-de-usuarios-em-redes-sociais-baseadas-em-localizacao/

Link Prediction in Social Networks using multiple information sources
Grant, National Council for Scientific and Technological Development (Brasilia, BR), 2013-08 to 2014-02. GRANT_NUMBER: 151836/2013-2.

Link prediction in complex networks using community structure information
Grant, National Council for Scientific and Technological Development (Brasilia, BR), 2011-08 to 2013-03. GRANT_NUMBER: 137353/2011-1.

Publications

Advancing Multi-step Mathematical Reasoning in Large Language Models Through Multi-layered Self-reflection with Auto-prompting
André de Souza Loureiro, Jorge Valverde-Rebaza, Julieta Noguez, David Escarcega, Ricardo Marcacini
doi.org/10.1007/978-3-032-06078-5_12

GENDER-BASED COMPARISON OF EVALUATIONS USING VISUO-HAPTIC SIMULATORS AND CHATGPT ACTIVITIES VERSUS TRADITIONAL LEARNING MODELS
Rosa Maria Guadalupe Garcia-Castelan, Andrés González-Nucamendi, Luis Jaime Neri-Vitela, Victor Francisco Robledo-Rella, Juana Julieta Noguez-Monroy, Jorge Carlos Valverde-Rebaza
doi.org/10.21125/iceri.2025.0682

LLM4Gov: A Privacy-Preserving Approach to Teacher-Student Fine-Tuning of Distilled LLMs for the Public Sector
Ricardo M. Marcacini, Jorge Carlos Valverde-Rebaza, Marcelo A. S. Turine, Brucce Neves Santos, Silvio Levcovitz, Solange O. Rezende
doi.org/10.5753/lasdigov.2025.9471

Can We Trust AI Chatbots to Teach University Physics? A Performance Comparison of AI Chatbots
Víctor Robledo-Rella, Andrés Gonzalez-Nucamendi, Luis Neri, Rosa M.G. García-Castelán, Julieta Noguez, Jorge Valverde-Rebaza
doi.org/10.1109/educon62633.2025.11016485

Skill-based employment taxonomy in the global IT industry 5.0
Jorge Valverde-Rebaza, Fabiana Rodrigues de Góes, Julieta Noguez, Nathalia C. Da Silva
doi.org/10.3389/feduc.2025.1418184

A comparative study of a new challenge-based learning model for engineering majors
V. Robledo-Rella, L. Neri, R. M. G. García-Castelán, A. Gonzalez-Nucamendi, J. Valverde-Rebaza, J. Noguez
doi.org/10.3389/feduc.2025.1545071

A FACTOR ANALYSIS OF STUDENT MOTIVATIONS OF THE TECNOLOGICO DE MONTERREY LEADERS OF TOMORROW SCHOLARSHIP
Andres Gonzalez-Nucamendi, Luis Jaime Neri-Vitela, Víctor Robledo-Rella, Rosa María Guadalupe García-Castelán, Julieta Noguez, Jorge Carlos Valverde-Rebaza
doi.org/10.21125/iceri.2024.0752

Impact of Demographic and Co-Curricular Factors on the Academic Success of Students from Low-Income Families: A Scholarship Program Study
Andres Gonzalez-Nucamendi, Luis Neri, Rosa M.G. García-Castelán, Víctor Robledo-Rella, Jorge Valverde-Rebaza, Julieta Noguez
doi.org/10.1109/fie61694.2024.10893057

Exploring Supervised and Unsupervised Learning Techniques to Detect Ground-Glass Opacities in CT Images for COVID-19
Sharon Quispe, Ingrid Arellano, Pedro Shiguihara, Jorge Valverde-Rebaza
doi.org/10.1109/andescon61840.2024.10755619

Advanced large language models and visualization tools for data analytics learning
Jorge Valverde-Rebaza, Aram González, Octavio Navarro-Hinojosa, Julieta Noguez
doi.org/10.3389/feduc.2024.1418006

Integrating YOLO and 3D U-Net for COVID-19 Diagnosis on Chest CT Scans
Jorge Valverde-Rebaza, Guilherme R. Andreis, Pedro Shiguihara, Sebastián Paucar, Leandro Y. Mano, Fabiana Góes, Julieta Noguez, Nathalia C. Da Silva
doi.org/10.1109/cbms61543.2024.00011

Information Management and Big Data
Jorge C. Valverde-Rebaza
doi.org/10.1007/978-3-031-04447-2

Information Management and Big Data
Jorge C. Valverde-Rebaza
doi.org/10.1007/978-3-030-76228-5

Information Management and Big Data
Jorge C. Valverde-Rebaza
doi.org/10.1007/978-3-030-46140-9

Exploiting Geographical Data to Improve Recommender Systems for Business Opportunities in Urban Areas
Vinícius Ferreira, Alan Valejo, Paola Valdivia, Jorge C. Valverde-Rebaza
doi.org/10.1109/bracis.2019.00105

ANEW for Spanish Twitter Sentiment Analysis Using Instance-Based Multi-label Learning Algorithms
Rodrigo Palomino, Carlos Meléndez, David Mauricio, Jorge Valverde-Rebaza
doi.org/10.1007/978-3-030-11680-4_6

Job recommendation based on job seeker skills: An empirical study
Valverde-Rebaza, J., Puma, R., Bustios, P., Silva, N.C.

The role of location and social strength for friendship prediction in location-based social networks
Valverde-Rebaza, J.C., Roche, M., Poncelet, P., Lopes, A.D.A.
doi.org/10.1016/j.ipm.2018.02.004

A survey of the applications of Bayesian networks in agriculture
Drury, B., Valverde-Rebaza, J., Moura, M.-F., de Andrade Lopes, A.
doi.org/10.1016/j.engappai.2017.07.003

Exploiting social and mobility patterns for friendship prediction in location-based social networks
Valverde-Rebaza, J., Roche, M., Poncelet, P., De Andrade Lopes, A.
doi.org/10.1109/ICPR.2016.7900016

Network sampling based on centrality measures for relational classification
Berton, L., Vega-Oliveros, D.A., Valverde-Rebaza, J., da Silva, A.T., Lopes, A.A.

RGCLI: Robust Graph that Considers Labeled Instances for Semi-Supervised Learning
Berton, L., de Paulo Faleiros, T., Valejo, A., Valverde-Rebaza, J., de Andrade Lopes, A.
doi.org/10.1016/j.neucom.2016.11.053

The impact of network sampling on relational classification
Berton, L., Vega-Oliveros, D.A., Valverde-Rebaza, J., Da Silva, A.T., De Andrade Lopes, A.

A Naïve Bayes model based on overlapping groups for link prediction in online social networks
Valverde-Rebaza, J., Valejo, A., Berton, L., De Paulo Faleiros, T., De Andrade Lopes, A.
doi.org/10.1145/2695664.2695719

Causation generalization through the identification of equivalent nodes in causal sparse graphs constructed from text using node similarity strategies
Drury, B., Valverde-Rebaza, J., De Andrade Lopes, A.

Lazy Multi-label Learning Algorithms Based on Mutuality Strategies
Cherman, E.A., Spolaôr, N., Valverde-Rebaza, J., Monard, M.C.
doi.org/10.1007/s10846-014-0144-4

Link prediction in graph construction for supervised and semi-supervised learning
Berton, L., Valverde-Rebaza, J., De Andrade Lopes, A.
doi.org/10.1109/IJCNN.2015.7280543

A multilevel approach for overlapping community detection
Valejo, A., Valverde-Rebaza, J., De Andrade Lopes, A.
doi.org/10.1109/BRACIS.2014.76

An Open Source Tool for Crowd-Sourcing the Manual Annotation of Texts
Jorge C. Valverde-Rebaza
doi.org/10.1007/978-3-319-09761-9_31

Identification of related Brazilian Portuguese verb groups using overlapping community detection
Valejo, A., Drury, B., Valverde-Rebaza, J., Lopes, A.A.
doi.org/10.1007/978-3-319-09761-9

Link prediction in online social networks using group information
Valverde-Rebaza, J.C., De Andrade Lopes, A.
doi.org/10.1007/978-3-319-09153-2_3

Multilevel refinement based on neighborhood similarity
Valejo, A., Valverde-Rebaza, J., Drury, B., De Andrade Lopes, A.
doi.org/10.1145/2628194.262822

Music genre classification using traditional and relational approaches
Valverde-Rebaza, J., Soriano, A., Berton, L., De Oliveira, M.C.F., De Andrade Lopes, A.
doi.org/10.1109/BRACIS.2014.54

Exploiting behaviors of communities of twitter users for link prediction
Valverde-Rebaza, J., De Andrade Lopes, A.
doi.org/10.1007/s13278-013-0142-8

Link prediction in complex networks based on cluster information
Valverde-Rebaza, J.C., De Andrade Lopes, A.

Multiple kernel learning based on local and nonlinear combinations
Calderón-Niquín, M., Valverde-Rebaza, J.
doi.org/10.1109/CLEI.2012.6427179

Structural link prediction using community information on Twitter
Valverde-Rebaza, J., De Andrade Lopes, A.
doi.org/10.1109/CASoN.2012.6412391

Service

Reviewer: Computing (journal), review activity in 2021.

Reviewer: Knowledge-Based Systems (journal), review activity in 2025.

Reviewer: Online Social Networks and Media (journal), review activity in 2024–2025.